

Introduction to Data Science 11372 & 11516

Assignment 1 - Data Wrangling - Solution

Ruhan Shafi

Your student ID (as recorded in Canvas): u3284342

Before you start

1. Replace uXXXXXXX in the `params:` block above with your own student ID, and put your name in the `author:` field. Everything in this document is generated from that ID.
2. Rename this file to `weatherdata_analysis_solution_uXXXXXX.Rmd`.
3. Work through the tasks in order. Each task has its own chunk or prose block below — write your answer in the block provided and do not renumber them.
4. Knit the document before you submit and check that every chunk ran.

This document knits with the `rmdformats` output format. If knitting fails with an error about `rmdformats`, install it once with `install.packages("rmdformats")` and knit again.

Setup — generate your dataset

Run this chunk exactly as provided. **Do not edit the generator.** If you change it, the dataset your marker regenerates will not match yours and your fingerprint check will fail.

```
knitr::opts_chunk$set(echo = TRUE)

## -----
## DO NOT EDIT THIS BLOCK
## -----

seed_from_id <- function(student_id) {
  id <- toupper(gsub("[^0-9A-Za-z]", "", as.character(student_id)))
  if (nchar(id) == 0L) stop("student_id must contain letters or digits.")
  ch <- utf8ToInt(id)
  h <- 5381 # djb2-style rolling hash
  for (k in ch) h <- (h * 33 + k) %% 2147483647
  as.integer(h %% 1000000000) + 1L
}

generate_weather_data <- function(student_id,
                                   start_date = as.Date("2024-01-01"),
                                   end_date   = as.Date("2025-12-31")) {
```

```

RNGkind("Mersenne-Twister", "Inversion", "Rejection") # reproducible on R >= 3.6
set.seed(seed_from_id(student_id))

dates <- seq.Date(start_date, end_date, by = "day")
n <- length(dates)
doy <- as.numeric(format(dates, "%j"))
yr <- as.integer(format(dates, "%Y"))
w <- 2 * pi * (doy - 20) / 365.25 # temperature peak ~20 January

## 1. Rainfall: two-state Markov chain (wet and dry spells) -----
seas_wet <- 1 + 0.18 * cos(2 * pi * (doy - 105) / 365.25) # autumn maximum
p_dw <- pmin(0.9, 0.260 * seas_wet) # P(wet today | dry yesterday)
p_ww <- pmin(0.9, 0.580 * seas_wet) # P(wet today | wet yesterday)
wet <- logical(n); wet[1] <- runif(1) < 0.3
for (i in 2:n) wet[i] <- runif(1) < if (wet[i - 1]) p_ww[i] else p_dw[i]
rain <- ifelse(wet, rgamma(n, shape = 0.80, scale = 10.5), 0)
rain <- round(pmin(rain, 210), 1)
rain[rain < 0.2] <- 0 # below gauge resolution
wet <- rain > 0

## 2. Temperature: climatology + AR(1) anomaly + summer heatwave spikes ---
t_clim <- 18.8 + 5.3 * cos(w)
phi <- 0.74; sig <- 2.6 * sqrt(1 - phi^2)
anom <- numeric(n); anom[1] <- rnorm(1, 0, 2.6)
for (i in 2:n) anom[i] <- phi * anom[i - 1] + rnorm(1, 0, sig)

spike <- rbinom(n, 1, 0.012) * rexp(n, 1 / 3.2) * pmax(0, cos(w))^1.5
heat <- spike # smear over a 3-day heatwave
heat[-1] <- heat[-1] + 0.70 * spike[-n]
heat[-(1:2)] <- heat[-(1:2)] + 0.40 * spike[-((n-1):n)]
anom <- anom + pmin(heat, 7)

temp <- t_clim + anom - 1.6 * wet

## 3. Relative humidity -----
rh <- 65 + 2 * cos(w) - 2.1 * anom + 10 * wet + 6 * pmin(1, rain / 20) +
  rnorm(n, 0, ifelse(wet, 4.5, 6.5))
rh <- pmin(99, pmax(24, rh))

## 4. Wind speed -----
wind <- rweibull(n, shape = 2.1, scale = 15) + 3.5 * wet +
  1.2 * cos(2 * pi * (doy - 300) / 365.25)
wind <- pmax(1.5, wind)

## 5. Solar exposure: clear-sky cycle x cloud factor -----
ws <- 2 * pi * (doy - 172) / 365.25 # minimum at winter solstice
clear <- 22 - 10.0 * cos(ws)
cf <- ifelse(wet, rbeta(n, 2.4, 2.6), rbeta(n, 9, 1.4))
solar <- pmax(0.6, clear * cf)

## 6. Soil moisture: single-layer bucket with memory -----
sm <- numeric(n); sm[1] <- 30
for (i in 2:n) {

```

```

et    <- (0.12 + 0.035 * max(temp[i], 0)) * (solar[i] / 18)  # evapotranspiration
drain <- 0.032 * (sm[i - 1] - 8)                               # slow drainage
sm[i] <- min(max(sm[i - 1] + 0.30 * rain[i] - et - drain, 8), 55)
}
sm <- pmin(56, pmax(5.5, sm + rnorm(n, 0, 0.35)))

## 7. Diurnal range and daily extremes -----
drange <- pmin(14, pmax(2.5, 6.0 + 3.2 * (solar / clear) - 0.03 * rh +
                        0.40 * anom + rnorm(n, 0, 0.8)))
tmax   <- temp + drange / 2

## 8. Fire danger: McArthur FFDI on afternoon conditions, binned 1-10 -----
drought <- 10 * (1 - (sm - 8) / (55 - 8))
rh_pm   <- pmax(8, rh - 18)
wind_pm <- wind * 1.25
ffdi    <- 2 * exp(-0.450 + 0.987 * log(pmax(drought, 0.1)) -
                  0.0345 * rh_pm + 0.0338 * tmax + 0.0234 * wind_pm)
fri     <- as.integer(cut(ffdi,
                          breaks = c(-Inf, 1, 2.5, 5, 8, 12, 17, 24, 35, 50, Inf),
                          labels = FALSE))

## 9. D'harawal seasons: onsets jittered +/- 10 days each year -----
season_tbl <- data.frame(
  Season      = c("Burran", "Marrai'gang", "Burrugin",
                  "Wiritjiribin", "Ngonungi", "Parra'dowee"),
  WeatherName = c("Gadalung Marool", "Bana'murrai'yung", "Tugarah Tuli",
                  "Tugarah Gunya'marri", "Murrai'yunggory", "Goray'murrai"),
  Plant       = c("Weetjellian (Lightwood) blooming",
                  "Lilly Pilly fruit ripening",
                  "Burringoa (Forest Red Gum) flowering",
                  "Marrai'uo (Gossamer Wattle) flowering",
                  "Miwa Gawaian (Waratah) blooming",
                  "Kai'arrewan (Coast Myall) blooming"),
  Animal      = c("Male kangaroos aggressive",
                  "Marrai'gang (quolls) calling for mates",
                  "Echidnas following mating lines",
                  "Lyrebirds building display mounds",
                  "Flying foxes gathering at dusk",
                  "Eels moving downriver to the sea"),
  stringsAsFactors = FALSE)

nominal    <- c(91, 152, 213, 244, 305)          # nominal onsets, day-of-year
season_idx <- integer(n)
for (y in unique(yr)) {
  b <- sort(nominal + sample(-10:10, 5, replace = TRUE))
  k <- yr == y
  season_idx[k] <- findInterval(doy[k], b) + 1L
}

notes <- c(
  "Meat is not eaten during the hottest days; fruit and tubers gathered.",
  "Warm cloaks prepared; groups move toward the coast.",
  "Shellfish are not taken until the boo'kerrikin blooms.",

```

```

"Gentle spring rains begin; lyrebird calls heard at dawn.",
"Season of ceremony and gathering.",
"Fish and eels are plentiful in the bays and estuaries.")

df <- data.frame(
  Date           = dates,
  StationID      = "SYD-CICS-01",
  Temperature    = round(temp, 1),
  MaxTemp        = round(tmax, 1),
  MinTemp        = round(temp - drange / 2, 1),
  Rainfall       = rain,
  Humidity        = round(rh, 1),
  WindSpeed      = round(wind, 1),
  SolarExposure  = round(solar, 1),
  SoilMoisture    = round(sm, 1),
  FireRiskIndex  = fri,
  Season         = season_tbl$Season[season_idx],
  SeasonWeatherName = season_tbl$WeatherName[season_idx],
  IndicatorPlant  = season_tbl$Plant[season_idx],
  IndicatorAnimal = season_tbl$Animal[season_idx],
  CommunityNote   = notes[season_idx],
  stringsAsFactors = FALSE)

df$Season <- factor(df$Season, levels = season_tbl$Season)
df
}

dataset_fingerprint <- function(df) {
  num <- unlist(df[vapply(df, is.numeric, logical(1))], use.names = FALSE)
  s <- as.integer(round(sum(num, na.rm = TRUE)))           %% 1000000L
  k <- as.integer(sum(utf8ToInt(paste0(as.character(df$Season),
                                     collapse = ""))))   %% 10000L
  d <- as.integer(which.max(df$MaxTemp) + 7 * which.max(df$Rainfall)) %% 1000L
  sprintf("WX-%06d-%04d-%03d", s, k, d)
}

## -----
## Generate YOUR dataset
## -----
weather_data <- generate_weather_data(params$studentID)

## Untouched copy of the data exactly as generated. Do not modify or delete it.
## The fingerprint must always be computed from this object, never from
## `weather_data`, because later parts of the assignment add columns to
## `weather_data` and that would change the fingerprint.
weather_data_orig <- weather_data

cat("Student ID  :", params$studentID, "\n")

## Student ID   : u3284342

cat("Seed        :", seed_from_id(params$studentID), "\n")

## Seed         : 12109058

```

```
cat("Rows x Cols :", nrow(weather_data), "x", ncol(weather_data), "\n")

## Rows x Cols : 731 x 16

cat("FINGERPRINT :", dataset_fingerprint(weather_data_orig), "\n")

## FINGERPRINT : WX-134679-6835-871

head(weather_data)
```

##	Date	StationID	Temperature	MaxTemp	MinTemp	Rainfall	Humidity
## 1	2024-01-01	SYD-CICS-01	23.7	27.4	20.0	0.0	74.7
## 2	2024-01-02	SYD-CICS-01	24.3	27.1	21.5	0.0	71.1
## 3	2024-01-03	SYD-CICS-01	21.0	23.7	18.3	2.3	79.7
## 4	2024-01-04	SYD-CICS-01	20.8	22.3	19.2	2.7	83.9
## 5	2024-01-05	SYD-CICS-01	23.5	26.3	20.8	0.0	79.8
## 6	2024-01-06	SYD-CICS-01	26.4	30.4	22.4	0.0	82.4

##	WindSpeed	SolarExposure	SoilMoisture	FireRiskIndex	Season	SeasonWeatherName
## 1	5.5	31.0	30.2	3	Burran	Gadalung Marool
## 2	8.8	22.3	27.8	3	Burran	Gadalung Marool
## 3	6.7	18.4	27.1	2	Burran	Gadalung Marool
## 4	31.0	10.0	26.9	3	Burran	Gadalung Marool
## 5	26.7	29.5	25.0	4	Burran	Gadalung Marool
## 6	9.5	31.2	22.2	3	Burran	Gadalung Marool

##	IndicatorPlant	IndicatorAnimal
## 1	Weetjellan (Lightwood) blooming	Male kangaroos aggressive
## 2	Weetjellan (Lightwood) blooming	Male kangaroos aggressive
## 3	Weetjellan (Lightwood) blooming	Male kangaroos aggressive
## 4	Weetjellan (Lightwood) blooming	Male kangaroos aggressive
## 5	Weetjellan (Lightwood) blooming	Male kangaroos aggressive
## 6	Weetjellan (Lightwood) blooming	Male kangaroos aggressive

##	CommunityNote
## 1	Meat is not eaten during the hottest days; fruit and tubers gathered.
## 2	Meat is not eaten during the hottest days; fruit and tubers gathered.
## 3	Meat is not eaten during the hottest days; fruit and tubers gathered.
## 4	Meat is not eaten during the hottest days; fruit and tubers gathered.
## 5	Meat is not eaten during the hottest days; fruit and tubers gathered.
## 6	Meat is not eaten during the hottest days; fruit and tubers gathered.

Part A: Data Understanding

Every claim in Part A must be supported by a number that appears in your own knitted output.

A.1 — Describe your dataset (max 120 words)

Write your description here. Cover (a) the measured meteorological variables, (b) the seasonal-calendar variables, and (c) two specific analyses this dataset supports. Include at least three numeric values from your own data.

This dataset contains 731 daily records (1 Jan 2024 to 31 Dec 2025) from one Sydney station, SYD-CICS-01. The measured variables are mean, maximum and minimum temperature, rainfall, humidity, wind speed, solar exposure, soil moisture and a 1–10 fire risk index. Mean temperature was 17.9 °C, the hottest maximum reached 37.4 °C, and 2512.4 mm of rain fell, with 433 dry days. Each day is also labelled with one of six D’harawal seasons, plus its weather name, indicator plant, indicator animal and a community note; Burran is

the most frequent (172 days). The data supports comparing rainfall and soil moisture across the D'harawal seasons, and examining how soil moisture, humidity and temperature drive fire risk.

A.2 — Summary statistics

```
#-----#
cat("Number of days      :", nrow(weather_data), "\n")

## Number of days      : 731
cat("Date range         :", format(range(weather_data$Date), "%d %b %Y"), sep = " ")

## Date range          : 01 Jan 2024 31 Dec 2025
cat("\nMean temperature (°C) :", round(mean(weather_data$Temperature), 1), "\n")

##
## Mean temperature (°C) : 17.9
cat("Highest max temp (°C) :", max(weather_data$MaxTemp), "\n")

## Highest max temp (°C) : 37.4
cat("Total rainfall (mm)   :", round(sum(weather_data$Rainfall), 1), "\n")

## Total rainfall (mm)   : 2512.4
cat("Dry days (0 mm)      :", sum(weather_data$Rainfall == 0), "\n\n")

## Dry days (0 mm)      : 433
cat("Days per season:\n")

## Days per season:
print(table(weather_data$Season))

##
##      Burran  Marrai'gang  Burrugin Wiritjiribin  Ngoonungi  Parra'dowee
##      172      134      129      44      131      121
#-----#
```

- The numeric variable with the largest ratio of standard deviation to mean is ..., which tells me that ...
- `summary()` treats `Season` differently from `Temperature` because ... To get a mean for it I would have to ...

A.3 — Histograms

```
#-----#
#-----#
```

A.4 — Read your own plots (max 150 words total)

- The shape of my `Temperature` histogram is ..., and the data were generated that way because ...
- My number of dry days is The height of the first bar of the histogram is The difference of ... days is made up of ...

- c. The season with the highest mean SoilMoisture is ... (value ...). It is / is not the same season with the highest mean Rainfall, because ...

Part B: Vectors, Matrices and Arrays

Throughout Part B, “month” means a calendar month within a specific year (2024-01 ... 2025-12), giving 24 groups.

B.1 — avg_temp_month

```
#-----#
month_id <- format(weather_data$Date, "%Y-%m") # 24 groups: 2024-01 ... 2025-12

avg_temp_month <- tapply(weather_data$Temperature, month_id, mean)
round(avg_temp_month, 2)

## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##    22.78    21.73    21.12    18.28    15.49    14.44    12.34    11.81    15.60    19.91
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##    20.12    22.67    23.67    23.32    20.48    16.78    15.78    13.08    11.21    10.86
## 2025-09 2025-10 2025-11 2025-12
##    15.62    19.46    20.53    23.04

cat("Max:", round(max(avg_temp_month), 2), "°C in", names(which.max(avg_temp_month)), "\n")

## Max: 23.67 °C in 2025-01
#-----#
```

The maximum of avg_temp_month is 23.67 °C in month January 2025 2025-01 .

B.2 — avg_rain_month, avg_hum_month, avg_soil_month

```
#-----#
avg_rain_month <- tapply(weather_data$Rainfall, month_id, mean)
avg_hum_month <- tapply(weather_data$Humidity, month_id, mean)
avg_soil_month <- tapply(weather_data$SoilMoisture, month_id, mean)

round(avg_rain_month, 2)

## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##    2.60    6.04    4.74    3.84    5.00    2.35    7.04    3.93    1.82    2.74
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##    0.53    5.79    3.15    2.43    3.55    2.85    3.51    2.62    3.15    2.08
## 2025-09 2025-10 2025-11 2025-12
##    3.70    2.55    1.99    4.32

round(avg_hum_month, 2)

## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##    73.15    76.51    71.32    71.84    70.74    66.11    70.39    70.87    66.26    64.67
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##    69.32    73.34    69.45    71.01    74.40    74.17    70.44    69.38    71.82    75.21
## 2025-09 2025-10 2025-11 2025-12
```

```
##    67.31    66.82    68.38    71.62
```

```
round(avg_soil_month, 2)
```

```
## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##    17.09    23.72    24.89    22.68    33.25    34.64    45.77    37.62    27.86    14.07
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##     8.78    19.81    12.50    11.65    17.50    16.25    24.65    23.18    29.82    24.00
## 2025-09 2025-10 2025-11 2025-12
##    25.29    12.35    11.12    13.43
```

```
#-----#
```

B.3 — Matrix M_month (24 x 4)

```
#-----#
```

```
M_month <- cbind(Temperature = avg_temp_month,
                 Rainfall     = avg_rain_month,
                 Humidity     = avg_hum_month,
                 SoilMoisture = avg_soil_month)

dim(M_month)
```

```
## [1] 24  4
```

```
round(M_month, 2)
```

```
##           Temperature Rainfall Humidity SoilMoisture
## 2024-01         22.78      2.60    73.15         17.09
## 2024-02         21.73      6.04    76.51         23.72
## 2024-03         21.12      4.74    71.32         24.89
## 2024-04         18.28      3.84    71.84         22.68
## 2024-05         15.49      5.00    70.74         33.25
## 2024-06         14.44      2.35    66.11         34.64
## 2024-07         12.34      7.04    70.39         45.77
## 2024-08         11.81      3.93    70.87         37.62
## 2024-09         15.60      1.82    66.26         27.86
## 2024-10         19.91      2.74    64.67         14.07
## 2024-11         20.12      0.53    69.32           8.78
## 2024-12         22.67      5.79    73.34         19.81
## 2025-01         23.67      3.15    69.45         12.50
## 2025-02         23.32      2.43    71.01         11.65
## 2025-03         20.48      3.55    74.40         17.50
## 2025-04         16.78      2.85    74.17         16.25
## 2025-05         15.78      3.51    70.44         24.65
## 2025-06         13.08      2.62    69.38         23.18
## 2025-07         11.21      3.15    71.82         29.82
## 2025-08         10.86      2.08    75.21         24.00
## 2025-09         15.62      3.70    67.31         25.29
## 2025-10         19.46      2.55    66.82         12.35
## 2025-11         20.53      1.99    68.38         11.12
## 2025-12         23.04      4.32    71.62         13.43
```



```
#-----#
```

B.4 — Matrix `M_season`

```
#-----#
```

```
#-----#
```

B.5 — Array `A_month` (year x month x variable)

```
#-----#
```

```
#-----#
```

The July `SolarExposure` values are ... (2024) and ... (2025).

B.6 — `t(M_month) %*% avg_temp_month`

```
#-----#
```

```
#-----#
```

The units of the first element are ..., and the result is not a meaningful summary of my data because ...

B.7 — `above_below()` and the `TempFlag` column

```
#-----#
```

```
#-----#
```

Total number of “a” days: ... Month with the highest proportion of “a” days: ... (proportion ...)

The total is not exactly half of 731 because ...

B.8 — Year-on-year comparison

```
#-----#
```

```
#-----#
```

Largest warming: ... (... °C). Largest cooling: ... (... °C).

Part C: Looping and Conditional Statements

Each chunk below must use an explicit loop or conditional as described, and must contain **one comment line in your own words saying *why* you structured the loop the way you did** — not what the line does.

C.1 — Mean `SoilMoisture` for each of the 24 months

```
#-----#
```

```
# WHY I structured the loop this way:
```

```
#-----#
```

Highest: ... (month ...). Lowest: ... (month ...).

C.2 — Days with Temperature above 26 °C

```
#-----#
```

```
# WHY I structured the conditional this way:
```

```
#-----#
```

Count: ... , which is ... % of the record.

C.3 — Mean Humidity on days below 15 °C

```
#-----#
```

```
# WHY I structured the loop this way:
```

```
#-----#
```

Mean humidity below 15 °C: ... ; all other days: The difference goes in that direction because ...

C.4 — FireRiskIndex above 4 in Burran and Burrugin

```
#-----#
```

```
# WHY I structured the conditional this way:
```

```
#-----#
```

Burran: ... days. Burrugin: ... days. Each count is / is not consistent with its D'harawal weather name because ...

C.5 — Total Rainfall matrix (24 months x 6 seasons)

```
#-----#
```

```
# WHY I structured the nested loops this way:
```

```
#-----#
```

- Cells that are exactly zero: ... out of 144.
- So many cells are zero because The pattern of the non-zero cells tells me that ...
- Month containing two seasons: ... ; transition date: ...

C.6 — Longest dry spell

```
#-----#
```

```
# WHY I structured the loop this way:
```

```
#-----#
```

Length: ... days. Start: End: SoilMoisture on the first day: ... ; on the last day: The change goes in that direction because ...

Part D: Reasoning, Prediction and Verification

D.1 — Explain your own code (max 150 words)

I am explaining my solution to [Part B task 7 / Part C task 5 / Part C task 6].

Step by step, ...

After each major step the object holds ...

The line that would break with a different number of rows is ..., because ...

D.2 — Predict, then check

Write your predictions in the chunk below **before running anything**.

```
# PREDICTION (write BEFORE running):  
# Q1. Roughly how many days in your record have Rainfall above 20 mm?  
#   My prediction: ____  
# Q2. Will the median of Rainfall be greater or less than the mean? By roughly how much?  
#   My prediction: ____  
# Q3. What proportion of the 731 days in your record are dry (Rainfall == 0)?  
#   My prediction: ____
```

Now run the three lines below and reconcile.

```
sum(weather_data$Rainfall > 20) # Q1
```

```
## [1] 28
```

```
c(mean = mean(weather_data$Rainfall),  
   median = median(weather_data$Rainfall)) # Q2
```

```
##      mean      median  
## 3.436936 0.000000
```

```
mean(weather_data$Rainfall == 0) # Q3
```

```
## [1] 0.5923393
```

Reconciliation (max 80 words):

Q1 actual ... — my prediction was wrong / right because ... Q2 actual ... — ... Q3 actual ... — ...

D.3 — Test four claims against your data

```
#-----#  
# Claim 1: wettest calendar month == month with highest mean soil moisture  
  
#-----#  
# Claim 2: 2025 warmer on average than 2024  
  
#-----#  
# Claim 3: fewer than 50 days with Temperature above 25 °C  
  
#-----#  
# Claim 4: longest run of consecutive dry days longer than 18 days
```

```
#-----#
```

Claim 1: numbers ... -> TRUE / FALSE Claim 2: numbers ... -> TRUE / FALSE Claim 3: numbers ...
-> TRUE / FALSE Claim 4: numbers ... -> TRUE / FALSE

Claim 1 can be false even though rainfall is the only source of soil moisture because ... (two sentences)

Claims 2 and 3 came out differently for different students because ... (one sentence)

D.4 — `classify_day()` with thresholds from your own distribution

```
#-----#
```

```
#-----#
```

Quantile cut-points used: ... and ... (from `quantile()`). Counts: Hot ... , Mild ... , Cool

A quantile-based threshold is better than a fixed one on another site's data because ...

D.5 — Debugging: code that runs but is wrong

Both lines use `weather_data_orig`, the unmodified data frame from the setup chunk.

```
# (i)
avg_temp <- sum(weather_data_orig$Temperature) / length(weather_data_orig)
avg_temp
```

```
## [1] 817.8813
```

```
# (ii) intended: the mean humidity on hot, dry days
hot_dry <- mean(weather_data_orig$Humidity[weather_data_orig$Temperature > 25 |
                                             weather_data_orig$Rainfall == 0])
hot_dry
```

```
## [1] 65.65905
```

Now compute the correct versions, and the number of days each version of line (ii) selects.

```
#-----#
```

```
#-----#
```

Line (i): wrong value ... ; correct value What R actually did: ... , and the mistake is silent because ...

Line (ii): wrong value ... (selects ... days); correct value ... (selects ... days). What R actually did: ... , and the mistake is silent because ...

D.6 — Interpretation grounded in your own results

```
#-----#
```

```
#-----#
```

a. Seasons ranked by mean Temperature, with the number of days each covers:

1. ... (... °C, ... days)
2. ...

3. ...
4. ...
5. ...
6. ...

- b. “Hot”: “Dry” tested as a rate: My dataset does / does not support both halves. A seasonal total would mislead here because ...
- c. My dataset cannot be used as evidence for or against the Parra’dowee note because ... The data that would be needed instead are ... (two to three sentences)

D.7 — Write your own classifier

```
#-----#
#-----#
```

Thresholds and the statistic justifying each: - Wet: ... because ... - Stormy: ... because ...

Counts: Wet ... , Dry ... , Stormy

Stormiest date: ... — Rainfall ... mm, WindSpeed ... km/h, MaxTemp ... °C.

One day I disagree with: ... , because I would change ...

Verification and Reproducibility

Results Summary Table

Fill in this table **manually** from your own output above. Rows 1-11 are checked automatically against your student ID; row 12 is marked by inspection. Leave no cell blank; write **NA** and explain if you could not compute one.

#	Quantity	Your value
1	Dataset fingerprint (from the setup chunk)	
2	Mean Temperature over the whole record (2 dp)	
3	Total Rainfall over the whole record (1 dp)	
4	Number of days with Rainfall == 0	
5	Date of the highest MaxTemp , and that value	
6	Season with the highest mean Temperature , and that mean (2 dp)	
7	Number of days with FireRiskIndex > 4 in Burran	
8	Length of the longest dry spell (Part C task 6)	
9	Start date of the longest dry spell (earliest run, if tied)	
10	<code>cor(SoilMoisture, FireRiskIndex)</code> (2 dp)	
11	Number of "a" days from Part B task 7	
12	Your two <code>classify_day()</code> quantile cut-points (D.4)	

Reproducibility appendix

Leave this chunk unchanged as the **last chunk** of your submission.

```
cat("Student ID :", params$studentID, "\n")
```

```
## Student ID : u3284342
```

```

cat("Fingerprint :", dataset_fingerprint(weather_data_orig), "\n")

## Fingerprint : WX-134679-6835-871

cat("Knitted at :", format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"), "\n")

## Knitted at : 2026-09-23 13:21:14 AEST

cat("Chunks run :", length(knitr::all_labels()), "\n\n")

## Chunks run : 49

print(sessionInfo()$R.version$version.string)

## [1] "R version 4.6.1 (2026-06-24)"

```

Declaration

I confirm that the dataset analysed in this submission was generated from my own student ID, that the values reported in the Results Summary Table are the values produced by the code in this document, and that all explanations and interpretations are my own work.

Signed: _____ Date: _____

Overall Conclusion

Your overall reflection about this assignment goes here ...